# Applied Human Computer Interaction

## Assignment#03

# 23K-2001

## BCS-6J

**Q1)**

The evaluation technique used here is Cognitive Walkthrough. It is a usability inspection method used to check how easily first-time users can learn an interface by performing tasks step-by-step.

Process:

- Evaluators define a user task and break it into steps.
- At each step, they ask questions such as:
  - Will the user know what to do?
  - Will the user notice the correct action?
  - Will the user understand the feedback?
- The evaluator predicts possible user difficulties.

Real-life example: Testing whether a new user can successfully order food on a newly designed delivery app without instructions.

Advantages:

- Good for evaluating learnability for beginners.
- Does not require real users.
- Helps identify navigation and understanding problems early.

Limitations:

- Time-consuming for large systems.
- Depends heavily on evaluator assumptions.
- Less effective for experienced-user behavior.

**Q2)**

Heuristic evaluation is a usability inspection method where experts examine an interface using established usability principles such as consistency, visibility of system status, feedback, and error prevention.

Real-life example: An expert reviewing an ATM interface to check if buttons, messages, and navigation are consistent and understandable.

Advantages:

- Quick and low-cost evaluation method.
- Can identify many usability issues early in development.
- Does not require large numbers of users.

Limitations:

- Results depend on evaluator expertise.
- Different evaluators may find different issues.
- Does not measure actual user performance or satisfaction.

## Q3)

Model-based evaluation uses mathematical or cognitive models to predict user performance without conducting real user testing.

Examples include:

- GOMS model
- Keystroke Level Model (KLM)

Real-life example: Predicting how long it takes a user to withdraw cash from an ATM using keyboard and menu operations.

Applications:

- Predicting task completion time
- Comparing interface designs
- Improving efficiency in professional systems

Advantages:

- Saves testing time and cost.
- Useful during early design stages.
- Produces measurable performance predictions.

Limitations:

- Requires expert knowledge to build models.
- May not reflect real human emotions or mistakes.
- Less suitable for complex or unpredictable tasks.

**Q4)**

| Feature | Cognitive Walkthrough | Heuristic Evaluation | Model-Based Evaluation |
|---|---|---|---|
| Main Focus | Learnability for new users | Checking usability principles | Predicting user performance |
| Performed By | Evaluators/designers | Usability experts | System/model experts |
| Uses Real Users | No | No | No |
| Basis | Step-by-step task analysis | Heuristic principles | Mathematical/cognitive models |
| Best For | First-time user tasks | General usability issues | Efficiency and time prediction |
| Example | Using a new mobile app | Reviewing website consistency | Predicting ATM task time |
| Main Limitation | Depends on assumptions | Depends on expert quality | May ignore real user behavior |